Mock Interview Edition

Feed Ranking: From a Prize Metric to an Ordering

16 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

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HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

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Every scenario, by part

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SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

Round 1 Framing and metrics 4

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02 A better error score and a worse ordering Metric 5
03 Can you combine the posts' scale figures? Estimation 7
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Round 2 Data and labels 9

05 Whose behavior is it? Labels 9
06 One popularity number, or several? Freshness 11
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SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the problem: proxies, objectives and the sizing

Why a prize metric was only a proxy, why a better error score can give a worse ordering, and which figures can be combined.

01 What did the prize buy, and what shipped?

FRAMING

SCENARIO

In 2006 Netflix offered 1 million dollars for a 10% improvement in rating prediction. Years later, a colleague says the winning ensemble must be running in production, and that the prize shows rating accuracy was the goal. Check both claims against the posts.

OPENING PROBE

What was the proxy?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How far did the pieces get?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What shipped?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Was accuracy the goal?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 6

02 A better error score and a worse ordering

METRIC

SCENARIO

Two rating models are compared on three titles that a member would truly rate A 4.5, B 4.0 and C 3.0. Model 1 predicts 4.0, 4.2 and 3.1. Model 2 predicts 5.0, 3.5 and 2.0. A manager picks the model with the better RMSE. Show what the choice does to the row.

OPENING PROBE

What are the errors?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What are the RMSE values?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What are the orders?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Why does that happen?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 7

03 Can you combine the posts' scale figures?

ESTIMATION

SCENARIO

Part 1 says subscribers streamed 2 billion hours in the last quarter of 2011, and Netflix has more than 23 million subscribers. Part 2 says several million stream plays a day. An analyst divides one by another. Check which divisions make sense.

OPENING PROBE

What is the daily viewing?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does hours per play give?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does that mean?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Which figures can be used?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

04 Size the scoring load

ESTIMATION

SCENARIO

A homepage has rows of titles, and each title in a row is scored for the member. The posts give no catalog size or number of rows. Make a stated assumption, size the load, and say where the real cost is.

OPENING PROBE

What are the assumptions?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the load?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Is the arithmetic the cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would change the picture?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

PART 2

Data and labels: signals, windows, households and exposure

What the posts list as inputs, why one popularity number is not enough, what an account is not, and how a shown title differs from a chosen one.

05 Whose behavior is it?

LABELS

SCENARIO

The ranking signal is mostly behavioral: plays, queue adds, searches and how members interacted with what was shown. A teammate treats each account's history as one person's taste. Explain the problem and what it means for labels.

OPENING PROBE

What do the posts list?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What do they say about households?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What follows for labels?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does an implicit signal not say?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 10

06 One popularity number, or several?

FRESHNESS

SCENARIO

A product owner asks for one all-time popularity score per title, which is simpler. The posts compute popularity over several windows and within groups. Show what the single number misses.

OPENING PROBE

What do the posts do?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does a weekly figure miss?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does a global number miss?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does it cost?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 11

07 Shown is not chosen

EXPOSURE

SCENARIO

The posts say Netflix logs presentations: what was recommended, where it was shown, and how the member interacted. A teammate trains on plays as if every title had the same chance to be played. Explain the bias.

OPENING PROBE

What does the log hold?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the bias?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How can the log help?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you check?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

08 Review the row ranker

CODE REVIEW

SCENARIO

A colleague writes the row ranker below and says it implements the posts' scoring, sorting and filtering. Every member gets nearly the same order, and rows sometimes come back with fewer titles than requested. Find the faults.

```
def rank_row(candidates, w1, w2, top_n):
    ranked = sorted(
        candidates,
        key=lambda c: w1 * c["plays_last_week"] + w2 * c["predicted_rating"],
        reverse=True,
    )[:top_n]
    return [c["title_id"] for c in ranked if not c["already_watched"]]
```

OPENING PROBE

Why is everyone's order alike?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What is the fix for the scales?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

Why short rows?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What else is missing?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

State the fix, and a test.

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

PART 3

The ranking function: weights, learning to rank and reading evidence

How much the ratio of two weights decides, how a ranking model is learned, what a chart without a metric can and cannot say, and a loop the posts do not discuss.

09 How much do the weights decide?

RANKING FUNCTION

SCENARIO

The posts' example ranking function is a weighted sum of popularity and predicted rating. A product manager asks you to pick the weights. Show with numbers how much they matter, and how Netflix says to set them.

OPENING PROBE

What is the function?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Work an example.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

When does A win?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How does Netflix say to set the weights?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 15

10 Which model learns the ranking?

MODELS

SCENARIO

A team has a linear ranking function and wants a stronger model. The post lists several families and warns about matching features and model. Advise them.

OPENING PROBE

What does the post name?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is the warning?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What can a linear function not do?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you do?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

11 Read the bar chart

EVIDENCE

SCENARIO

Part 2 has a bar chart titled Ranking improvement over baseline. Popularity is the baseline at zero, adding ratings reaches about a third, and adding more features and optimized models reaches about 225%. A slide says Netflix improved ranking by 225%. Say what the chart supports.

OPENING PROBE

What does it show?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does it leave unsaid?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does it support?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How would you word the slide?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

12 Popular titles get more popular

RISK

SCENARIO

The ranking function includes popularity as an input, and popularity is computed from plays, which the ranking itself influences. A reviewer asks whether that is a problem. The posts do not discuss it.

OPENING PROBE

What is the loop?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Put numbers on two rounds.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What do the posts' tools offer?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you add?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

PART 4

Evaluation and operations: offline metrics, A/B tests and serving

Computing ranking metrics by hand, what a test with thousands of members can see, what to cache, and what grows at ten times the scale.

13 Compute the ranking metrics

METRICS

SCENARIO

The posts list NDCG, mean reciprocal rank and fraction of concordant pairs among the offline metrics. An interviewer asks you to compute two of them on small examples, and to say how far to trust them.

OPENING PROBE

MRR for three requests.

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

FCP for three titles.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What about the rest of the list?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How far to trust them?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

14 What can a test with thousands of members see?

EXPERIMENTS

SCENARIO

The posts say tests usually have thousands of members and 2 to 20 cells, and that scores of tests run in parallel, with engagement and retention as the trusted metrics. A colleague launches a 20-cell test and ships whichever cell has a significant gain. Examine it.

OPENING PROBE

How precise is a retention number?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What about a difference between two cells?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does a 20-cell test do?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What do the posts say?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 20

15 What can be cached?

SERVING

SCENARIO

A request must produce an ordered row for one member in real time. The posts say ranking is scoring, sorting and filtering, and give no architecture. Decide what to cache and what to compute.

OPENING PROBE

What is personal?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What is shared?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is the cost of caching it?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What is the serving path?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 21

16 Risks and ten times the scale

RISK

SCENARIO

A launch review asks what can go wrong with this ranking and what grows at ten times the members and plays. Use the posts and the sizing.

OPENING PROBE

What can go wrong?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does ten times do to scoring?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does it do to popularity?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does it do to tests?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 22

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	What did the prize buy, and what shipped?	5			
02	A better error score and a worse ordering	5			
03	Can you combine the posts' scale figures?	5			
04	Size the scoring load	5			
05	Whose behavior is it?	5			
06	One popularity number, or several?	5			
07	Shown is not chosen	5			
08	Review the row ranker	6			
09	How much do the weights decide?	5			
10	Which model learns the ranking?	5			
11	Read the bar chart	5			
12	Popular titles get more popular	5			
13	Compute the ranking metrics	5			
14	What can a test with thousands of members see?	5			
15	What can be cached?	5			
16	Risks and ten times the scale	5			